Part B — Off-Market Prediction

SSU Data Dash 2026 | Model development and measured results

Histogram gradient boosting achieved 9.08% MdAPE and 11.60% MAPE over 10,069 held-out sales. This is 49.2% lower MdAPE than the rolling community/type baseline (17.89%).

Model and information boundary

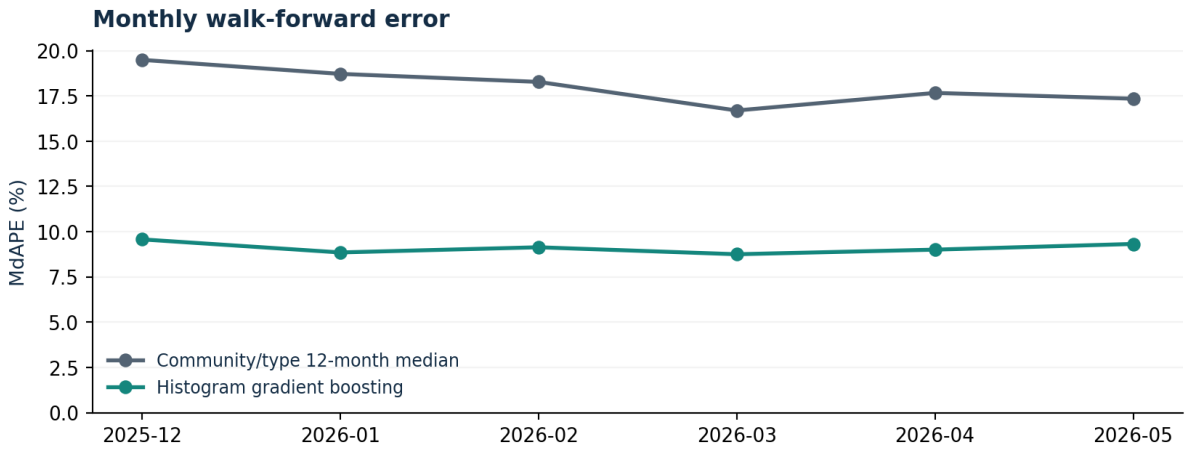
Predict log(sold_price), then exponentiate to dollars. Inputs cover size and room counts, property type/style, synthetic location, lot dimensions with missingness indicators, and listing-date calendar features, plus fixed remarks flags for condition, amenities and layout. Area ranges are represented by their midpoint and width. Histogram gradient boosting builds trees sequentially; category encoding and numeric imputation are fitted within each training fold.

No list_price, sold_price, date_sold, days_on_market, external_avm_estimate, external_avm_estimate_date or ml_num enters the learned feature matrix. sold_price is only the target; date_sold only defines training cutoffs and evaluation groups. No external data is used.

Untouched six-month walk-forward backtest

Tune on October–November 2025 only, selecting the lowest pooled MdAPE (MAPE breaks ties). Freeze the model configuration before December 2025–May 2026. For each of those six months, refit on sales strictly before the month's start and predict that month's held-out sales. This follows the official sale-month backtest; it is not a strict per-home listing-time simulation.

Model	MdAPE	MAPE	Homes
Histogram gradient boosting	9.08%	11.60%	10,069
Community/type 12-month median	17.89%	22.25%	10,069
10 nearest earlier sales	20.84%	27.43%	10,069
Global median	38.46%	46.68%	10,069



MdAPE = median(100 × |prediction − sale| / sale); MAPE uses the mean. Both are percentages of actual sale price, and lower is better. MdAPE is the competition's primary metric.

Selection, final model and limits

Earlier tuning: October–November 2025

The table below is used only for model selection. It is deliberately separated from the later six-month backtest.

Model	MdAPE	MAPE	Homes
Histogram gradient boosting	9.31%	11.71%	4,117
CatBoost depth 7 / RMSE	9.54%	12.07%	4,117
CatBoost depth 7 / MAE	9.58%	12.03%	4,117
CatBoost depth 6 / RMSE	9.94%	12.64%	4,117
CatBoost depth 6 / no remarks	10.51%	13.35%	4,117
Random forest	13.28%	16.99%	4,117
Community/type 12-month median	19.02%	23.58%	4,117
10 nearest earlier sales	21.59%	28.88%	4,117
Global median	35.48%	43.24%	4,117

Frozen selected configuration

max_iter=350; learning_rate=0.07; max_leaf_nodes=31; l2_regularization=10.0; remarks features=on; log-price target; random seed 42.

Interpretation and deliverable

The selected model puts 53.8% of held-out predictions within 10% of sale price and 83.8% within 20%. Median signed error: +0.49%. The final model is refitted on all 150,000 training sales, then predicts the supplied off-market test homes.

The $2m+ group had 14.68% MdAPE across 719 homes, with -9.57% median signed error. The data is synthetic; results do not establish real-world appraisal accuracy. Six months do not capture every future market regime. Expensive or unusual homes may have larger errors. The June–September 2026 test prices are hidden, so test accuracy is unknown. Do not describe the validation score as a hidden-test score.

Report use and provenance

This standalone Part B handoff is not the complete team submission. Insert report_section.md into the team's four-page report and add member names, GitHub and video links. The Part B video allocation is approximately 75 seconds.

Official handout/starter: https://github.com/erwinzhang7/SSU-Data-Dash. Reference commit: 6c5831e12f523d617eccd1418d46a5cf2a0bb8b4. No external data was used.